Visveswaran V

Data Analyst



SUMMARY

Data Analyst with expertise in **SQL, Python, Power BI, and causal inference**, delivering actionable business intelligence from complex datasets. Proven track record of **reducing algorithmic bias by 45%** in credit-scoring models and **uncovering causal insights** that overturned company assumptions on customer churn. Built **FairViT-Med**, a fairness-aware Vision Transformer trained on **112,000+ medical images**, cutting demographic AUC gaps by **64%** while deploying containerized **FastAPI** services with **MLflow** experiment tracking. Google Data Analytics Certified and **NPTEL Gold Medalist**, skilled in **predictive modeling, EDA, data visualization, and fairness-aware AI**. Seeking to leverage analytical precision and stakeholder-ready storytelling in a **Data Analyst** role to drive measurable business impact.

PORTFOLIO PROJECTS

RetailIQ — Causal Marketing Impact & Customer Lifetime Value System Jul – Sep 2026
Independent Portfolio Project | SQL, Python, XGBoost, SHAP, DoWhy, Power BI, Excel

- Engineered an end-to-end causal analytics system on ~99,000 Olist e-commerce orders, combining SQL data modeling, RFM segmentation, and an XGBoost churn/CLV model (ROC-AUC 0.746).
- Applied DoWhy causal graphs and propensity-score matching to reverse a naïve correlation — vouchers were actually linked to a 17.5-point rise in churn — then translated the finding into a Power BI dashboard and Excel ROI model flagging an 18,000-customer high-risk segment.

Portfolio Project — Business & Retail Analytics Case Studies Jan 2024 – Present
Self Analytics | Python, SQL, Power BI, Excel

- Produced 10+ end-to-end case studies (Swiggy, Zomato, Netflix, Amazon), analyzing 5+ years of sales and customer-behavior data through exploratory data analysis, regression, and trend decomposition.
- Designed 10+ Power BI dashboards and Excel models that trimmed data-interpretation time by 40%, after cleaning and preprocessing datasets of up to 500,000+ records with Pandas and NumPy.

Causal Fairness-Aware Credit Scoring March 2026
Research Project | Python, DoWhy, SHAP, Streamlit

- Developed a responsible-AI credit risk pipeline on the UCI German Credit Dataset, lowering the algorithmic bias parity gap by 45% across 3 debiasing strategies while sustaining accuracy above 65%.
- Implemented a SHAP explainability layer and deployed a Streamlit app with real-time scoring, a live fairness dashboard, and a DoWhy-based counterfactual scenario explorer.

FairViT-Med: Fairness-Aware Vision Transformer for Medical Diagnosis Dec 2025 – Mar 2026
Research Project | PyTorch, ViT-B/16, MLflow, FastAPI, Docker

- Trained a ViT-B/16 Vision Transformer on 112,120 NIH Chest X-ray images, cutting the sex-stratified AUC fairness gap by 64% (0.061 to 0.022) via reweighted loss functions and balanced sampling while holding overall AUC at 0.553.
- Optimized a 43GB medical-imaging preprocessing pipeline on local GPU hardware using PyTorch DataLoaders with parallel workers and mixed-precision training, shrinking per-epoch load time by 35%.
- Shipped the fairness-mitigated model as a containerized FastAPI service with Docker and tracked every experiment in MLflow, alongside a 100+ page reproducible research report.

CONTACT

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EDUCATION

B.Tech, Artificial Intelligence & Data Science
Gnanamani College of Technology, Namakkal, Tamil Nadu
2023 – 2027 (Expected) | CGPA: 7.7/10.0

Higher Secondary Education (12th)
Government Higher Secondary School,
Vembadithalam, Salem
Percentage: 64%

SKILLS

Programming
- Python, SQL, R, C

Machine Learning & AI
- Scikit-learn, PyTorch, XGBoost
- SHAP, DoWhy, Causal Inference

Analytics & BI
- Power BI, Tableau, Advanced Excel
- Data Visualization, Data Storytelling
- Pandas, NumPy, Statistical Analysis

Tools & Platforms
- Streamlit, FastAPI, Git/GitHub
- AWS S3, Jupyter, Google Colab

Languages
- English (Professional), Tamil (Native)

CERTIFICATIONS

- Google Data Analytics Professional Certificate — Google/Coursera
- NPTEL Gold Medal, Human-Computer Interaction (95/100) — IIT/NPTEL
- Generative AI: Introduction & Applications — IBM

HONORS & AWARDS

- 1st Prize, Technical Quiz — CYBORG-24, KIOT Salem (2024)
- 2nd Prize, Quiz Mania — Arivoli 2K26, KIOT Kakapalayam (2026)
- 2nd Prize, Proverb Competition — Arivoli 2K26, KIOT Kakapalayam (2026)
- 3rd Place, Design Dynasty & 3D Modeling Competition — KIOT Salem (2025)
- Startup TN Smart Card Holder — Govt. of Tamil Nadu Startup Ecosystem